EE-170L Computer Programming Lab

Laboratory Session 04

Switch / Break / Continue Statements

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1. Objective

The objective of this lab session is to understand and use the **switch statement in C++**. Students also learn how **break statements control the flow of execution** and how **continue works inside loops**. This lab also helps students practice creating **simple menu-driven programs**.

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2. Theory

The **switch statement** in C++ is used when a program needs to select one option from many possible choices based on the value of a variable. It is commonly used instead of long **if-else statements** when checking a single variable.

Each **case** represents a possible value of the variable. When the value matches a case, the code inside that case executes.

The **break statement** is used to stop execution of the switch statement once a matching case has been executed. If the break statement is not used, the program will continue executing the following cases.

The **default case** is optional and executes when none of the case values match the given input.

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3. Task 1 – Day Name using Switch

Write a C++ program that asks the user to enter a number between **1 and 7** and displays the corresponding **day of the week**.

```
#include <iostream>
using namespace std;

int main()
{
    int day;

    cout << "Enter day number (1-7): ";
    cin >> day;

    switch(day)
    {
        case 1:
            cout << "Monday";
            break;

        case 2:
            cout << "Tuesday";
            break;

        case 3:
            cout << "Wednesday";
            break;

        case 4:
            cout << "Thursday";
            break;

        case 5:
            cout << "Friday";
            break;

        case 6:
            cout << "Saturday";
            break;

        case 7:
            cout << "Sunday";
            break;

        default:
            cout << "Invalid day number";
    }

    return 0;
}
```

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4. Task 2 – Grade Evaluation using Switch

Write a program that takes a **character grade** from the user and prints the appropriate message.

```
#include <iostream>
using namespace std;
```

```
int main()
{
    char grade;

    cout << "Enter grade (A, B, C, D, F): ";
    cin >> grade;

    switch(grade)
    {
        case 'A':
            cout << "Excellent";
            break;

        case 'B':
            cout << "Very Good";
            break;

        case 'C':
            cout << "Good";
            break;

        case 'D':
            cout << "Pass";
            break;

        case 'F':
            cout << "Fail";
            break;

        default:
            cout << "Invalid Grade";
    }

    return 0;
}
```

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5. Task 3 – Unit Conversion Menu Program

Write a program that displays a menu and performs unit conversions based on the user's choice.

```
#include <iostream>
using namespace std;

int main()
{
    int choice;
    float value;

    cout << "1 -> Convert Kilometers to Meters" << endl;
    cout << "2 -> Convert Meters to Centimeters" << endl;
    cout << "3 -> Convert Kilograms to Grams" << endl;
    cout << "4 -> Convert Celsius to Fahrenheit" << endl;
```

```
    cout << "Enter your choice: ";
    cin >> choice;

    cout << "Enter value: ";
    cin >> value;

    switch(choice)
    {
        case 1:
            cout << "Meters = " << value * 1000;
            break;

        case 2:
            cout << "Centimeters = " << value * 100;
            break;

        case 3:
            cout << "Grams = " << value * 1000;
            break;

        case 4:
            cout << "Fahrenheit = " << (value * 9/5) + 32;
            break;

        default:
            cout << "Invalid choice";
    }

    return 0;
}
```